Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*A fermentation and extrusion technology for sustainable alternative seafood products from yeast biomass.*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio·Medical		*Bio·Medical*	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial Biotechnology	*Bio-process Technologies*	*Industrial Biotechnology*	*Functional Food Ingredients*
Concept of Proposed Technology	*The proposed technology enables the production of protein-rich, textured alternative seafood ingredients based on the biomass of oleaginous yeast. The yeast is cultivated via fermentation using lipid-rich side streams such as used cooking oils, enabling a resource-efficient upcycling of low-value residues into food-grade biomass. The harvested biomass is structured by extrusion to obtain fibrous textures comparable to conventional fish products. By adjusting fermentation conditions, formulation, and extrusion parameters, sensory and functional properties can be tailored to different seafood applications (e.g., white fish-like or salmon-like products).* *The result is a scalable, sustainable technology platform that combines microbial biotechnology with industrial food processing for alternative (vegan) seafood.*			

Rationale for Joint R&D Project	*Initial laboratory feasibility has demonstrated (i) robust yeast growth on lipid side streams, (ii) lab-scale wet/dry extrusion for texture formation, and (iii) first sensory indications (umami/fish-like notes) through natural flavors. To move towards industrial robustness and scale, joint R&D is required to:* • *stabilize process windows (fermentation → biomass → extrusion),* • *handle raw material variability (different side-stream oils),* • *define application-specific KPIs and quality standards, and* • *pilot the process under industry-relevant conditions.* *Collaboration between the Fraunhofer IGB and Korean SMEs is essential, as the Fraunhofer IGB contributes microbial bioprocessing, analytics and extrusion know-how, while Korean SMEs provide application development, pilot processing, regulatory and market feedback, accelerating industrial validation and reducing time-to-market. Given Korea's strong fermentation heritage (kimchi, jang, gochujang) and modern industrial capabilities in fermented foods, Korean consumers and SMEs are structurally receptive to fermentation-enabled innovations. This cultural and market fit reduces adoption risk and strengthens the case for joint development and industrial validation of a yeast-based alternative seafood platform.*
Key Words	1. *Alternative (vegan) seafood* 2. *Yeast fermentation* 3. *Extrusion processing* 4. *Sustainable protein production* 5. *Waste stream recycling*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3 - 4)	End of the Project	Level (6)
R&D Objectives	***Objective 1 – Fermentation Optimization:*** *Develop robust yeast cultivation on lipid-based side streams with consistent biomass yield and composition.* ***Objective 2 – Texturing & Formulation:*** *Establish extrusion and formulation strategies to achieve reproducible fibrous textures tailored to seafood-like applications.* ***Objective 3 – Process Integration & Scale-up:*** *Define and validate end-to-end process windows (fermentation → biomass handling → extrusion) at pilot scale.* ***Objective 4 – Application KPIs & Validation:*** *Define product KPIs (texture, sensory, nutrition) and verify industrial compatibility (e.g., further processing, storage stability).*			
R&D Target Specifications	• *Biomass protein content (dry basis): ≥45%* • *Extruded texture robustness: ≥90% batch-to-batch reproducibility of texture metrics (e.g., shear force / TPA firmness within ±10%)* • *Moisture window during extrusion: 48–55% (wet extrusion) with stable die temperature (e.g., 70–95°C window; exact setpoints to be optimized)* • *Off-flavor control: sensory panel scores within target range for fish-like profile; absence of undesirable notes* • *Process economics (pilot indicative): continuous operation at >70% on-stream time over >100 h pilot runs without quality drift*			
R&D Plan	***Phase I – Pre-study (6 months)*** *T1 – Feedstock & Variability: Assessment of 2–3 representative side-stream oil qualities; mapping impact on growth/yield/composition.* *T2 – Fermentation Window: Optimization of key parameters (C/N ratio, temperature, aeration) for stable biomass and target composition (2 L Lab scale)* *T3 – Extrusion & Formulation Screening: DoE-based screening of moisture, temperature profile, screw speed, plus simple natural flavor systems and definition of first texture KPIs.* *T4 – KPI Definition & Pilot Readiness: Consolidation of target specs, definition of QA/analytics, and preparation pilot run protocol for Phase II.* ***Milestones (Phase I):*** *M1 (Month 3): Defined fermentation window with ≥2 qualified feedstocks* *M2 (Month 6): Reproducible lab-scale texture meeting preliminary KPIs, Phase II plan finalized.* ***Phase II – Main R&D (30 months)*** *WP1 – Pilot Fermentation: Scale up to pilot fermenters (minimum 1 m^3), validation of consistency and run stability.* *WP2 – Continuous Extrusion: Establishment of continuous wet extrusion with in-line monitoring, verification of reproducibility at pilot scale.* *WP3 – Application Development: Tailoring of textures for two target product profiles (e.g., white-fish-like / salmon-like), validation of secondary processing (forming/breading, thermal steps).* *WP4 – Product & Process Validation: Shelf-life indicative tests, HACCP concept, preliminary cost model; definition of industrial integration options.* *WP5 – Demo & Handover: Production of pilot batches; compile technology package (SOPs, specs, QA).* ***Milestones (Phase II):***			

	M3 (≈ Month 12 of Phase II): Stable pilot fermentation window confirmed *M4 (≈ Month 20): Continuous extrusion reproducibility >90 % on texture KPIs* *M5 (≈ Month 28–30): Pilot product meeting application specs, industrial validation dossier compiled.*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*The Fraunhofer IGB will be responsible for the development, optimization, and validation of the fermentation- and extrusion-based technology, with a focus on process robustness, scalability, and industrial relevance.* *At the beginning of the project, Fraunhofer will perform systematic strain and process screening at small scale using high-throughput cultivation systems such as BioLector® XT. These experiments will be used to evaluate growth behavior, substrate utilization, and biomass composition under varying conditions, including different lipid-based side streams, nutrient ratios, temperature, and aeration. This approach enables rapid identification of robust operating windows and early assessment of feedstock-related variability.* *Based on the screening results, selected conditions will be transferred to bench-scale stirred tank fermenters (e.g. 2-liter scale). At this stage, Fraunhofer will focus on process transferability and scale-relevant parameters, including oxygen transfer, mixing, pH control, and reproducibility of biomass yield and quality. The goal is to establish scalable and industry-oriented fermentation strategies rather than laboratory-specific optima.* *Fraunhofer will further conduct analytical characterization of the produced biomass, including protein content, composition, and functional properties relevant for downstream processing. These data will be used to define key performance indicators (KPIs) and quality requirements for subsequent extrusion steps.* *In addition, the institute will develop and optimize extrusion-based structuring processes, investigating moisture content, temperature profiles, and processing conditions to achieve reproducible fibrous textures suitable for alternative seafood applications. Fermentation and extrusion results will be integrated to ensure end-to-end process compatibility.* *Overall, Fraunhofer will provide an evidence-based and industrially applicable process foundation, reducing technical risks and enabling Korean SME partners to focus on application development, pilot-scale validation, and commercialization.*
Expected R&R of Korean SME Partner	*The Korean SME partner will contribute application-driven expertise and industrial perspective to support the transition of the technology from laboratory scale to pilot-scale and market-relevant validation.* *The SME will define target product profiles and performance requirements for alternative seafood applications, including texture, processing behavior, and application-specific constraints. This input will guide process optimization and ensure alignment with industrial and market needs.* *Furthermore, the Korean SME will support pilot-scale processing and validation activities, including trials of fermentation outputs and extruded materials under industry-relevant conditions. Feedback on manufacturability, handling, and compatibility with existing food processing lines will be provided.* *In addition, the SME will contribute to the assessment of downstream processing, shelf-life, and preliminary regulatory considerations in the Korean market. Through continuous feedback, the SME partner will help shape a commercially viable and scalable technology pathway towards Phase II and beyond.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The project will deliver a validated, end-to-end process for producing yeast-based alternative seafood ingredients that integrates fermentation, biomass handling and wet extrusion into a reproducible, pilot-ready workflow. Building on BioLector® XT screenings and their targeted transfer to 2-liter stirred-tank fermenters, we will define robust operating windows for media composition, oxygen transfer, pH and moisture management, and establish analytical and quality-assurance methods, such as biomass composition profiling and texture profile analysis—that ensure batch-to-batch consistency. By Phase I completion, application-specific performance targets (texture, sensory profile and processing behavior) will be consolidated into clear KPIs alongside a preliminary manufacturability and cost view, forming a strong foundation for Phase II scale-up and industrial validation.* *Commercially, the results constitute a platform technology that Korean SMEs can integrate into existing food-processing lines with minimal adaptation, thus shortening time-to-market for alternative seafood products. The technology enables the conversion of lipid-rich side streams, such as used cooking oils, into higher-value, food-grade biomass and thereby opens new business opportunities across the value chain: from side-stream collection and preprocessing to ingredient production and final product formulation. The documentation, specifications and SOPs developed during the project will support IP generation and de-risked technology transfer or licensing models tailored to SME needs.* *Socio-economically, the project strengthens SME competitiveness in the fast-growing alternative protein space and supports the creation of high-skilled jobs in food biotechnology, process engineering and quality control. By diversifying protein sources through controlled microbial fermentation, it contributes to food system resilience and helps stabilize supply independent of fluctuations in capture fisheries and aquaculture. Localized upcycling of side streams fosters regional circular economies, keeping value within domestic supply chains, while joint R&D builds lasting capabilities in bioprocessing and extrusion, both within the consortium and the broader SME ecosystem.* *From a sustainability perspective, the technology advances resource efficiency and circularity by converting low-value residues into edible protein ingredients, with the potential to reduce land use, freshwater demand and greenhouse-gas emissions compared to conventional seafood. Phase II will further quantify these benefits through targeted process optimization and indicative LCA activities, while continuous operation strategies aim to reduce specific energy demand and achieve stable on-stream times in pilot runs. By offering market-ready alternatives, the project can also help alleviate pressure on wild fisheries and support marine biodiversity protection.* *Finally, the outcomes are aligned with regulatory readiness and standards. The use of non-GMO yeast biomass with an established food-safety track record facilitates the preparation of HACCP concepts and streamlines alignment with Korean food regulations. The project will deliver specification sheets, analytical methods and traceability concepts that support certification, customer audits and ESG reporting, ensuring that Korean SMEs can adopt the technology with confidence and progress efficiently toward commercialization.*

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization